UBER EATS OPERATIONAL & CUSTOMER INTELLIGENCE

End-to-End ML Pipeline: Findings & Recommendations

1. Executive summary

We combined four analytical models into one pipeline covering customer and restaurant segmentation, review sentiment, delivery time prediction and demand forecasting. The data covers 5,000 orders, 500 customers, 50 restaurants and 100 drivers across five cities. The headline findings:

₹35.8 lakh	60 min	11.8%	14%
order value delivered (4,410 orders)	average delivery time; 49% of orders take over an hour	orders cancelled or failed (₹4.6 lakh of order value)	of reviews are negative

- **Two customer groups matter most.** Premium Frequent Diners bring 41% of order value. Weekend High-Ticket customers spend the most per order but leave negative reviews at nearly three times the rate of everyone else (29% vs 10%).
- **Delivery delays are mostly about distance, kitchen speed and conditions.** Each extra kilometre adds about 3 minutes and each extra minute in the kitchen adds about 1 minute. Heavy traffic adds about 16 minutes and rain about 12 minutes.
- **The 20 slow-preparation restaurants are the top intervention target.** They handle 42% of orders, take 68 minutes on average to deliver, and 68% of their orders exceed an hour.
- **Demand is steady, not spiky.** Orders run at about 1.1 per hour with no statistically reliable peak by hour or weekday, so time-of-day staffing is not supported by this data. Delhi and Mumbai carry 53% of volume.
- **Tips are not influenced by service quality.** Tip size follows order value and a tip percentage that customers appear to pick independently of speed, weather or traffic.

2. Model comparison summary

Each part of the pipeline tested more than one approach and kept the best performer. The table shows what was tested, what was chosen, and what the result means in plain terms.

Area	Approaches tested	Selected	Result in plain terms
Customer segments	K-Means, DBSCAN	K-Means (4 groups)	Groups are real but overlap: separation score 0.16 out of 1. Use as behavioural tendencies, not hard categories.
Restaurant segments	K-Means, DBSCAN	K-Means (3 groups)	Moderate separation (0.26). The three groups are clearly different on prep time and rating.
Review sentiment	Naive Bayes, Linear SVM	Naive Bayes	Both classify 100% of test reviews correctly. This is inflated because reviews repeat a small set of phrases and sentiment is taken from star ratings.

Area	Approaches tested	Selected	Result in plain terms
Delivery time	Average-only baseline, Ridge, Lasso, Random Forest	Ridge Regression	Typical error of 3.2 minutes versus 14.0 for the baseline. Explains 94% of the variation in delivery time.
Hourly demand	Repeat last week (168 hours), flat average	Flat average	Repeating last week is less accurate (error 1.23 orders/hour) than a flat average (0.86), confirming there is no dependable weekly pattern.

Delivery time models in detail

Model	Typical error (min)	Larger-error measure (min)	Variation explained	Verdict
Ridge Regression	**3.23**	**4.11**	**94.4%**	**Selected**
Lasso Regression	3.24	4.12	94.3%	Equivalent
Random Forest	3.60	4.56	93.1%	Slightly weaker; fits training data more closely than new data
Baseline (average)	14.03	17.31	0%	Reference point

Ridge and Lasso perform almost identically, so the simpler, more explainable model was chosen. Results held up under five-fold cross-validation.

3. Business insights

Which customer segments are most valuable?

Answer: Premium Frequent Diners are the core. Weekend High-Ticket customers are valuable but at risk.

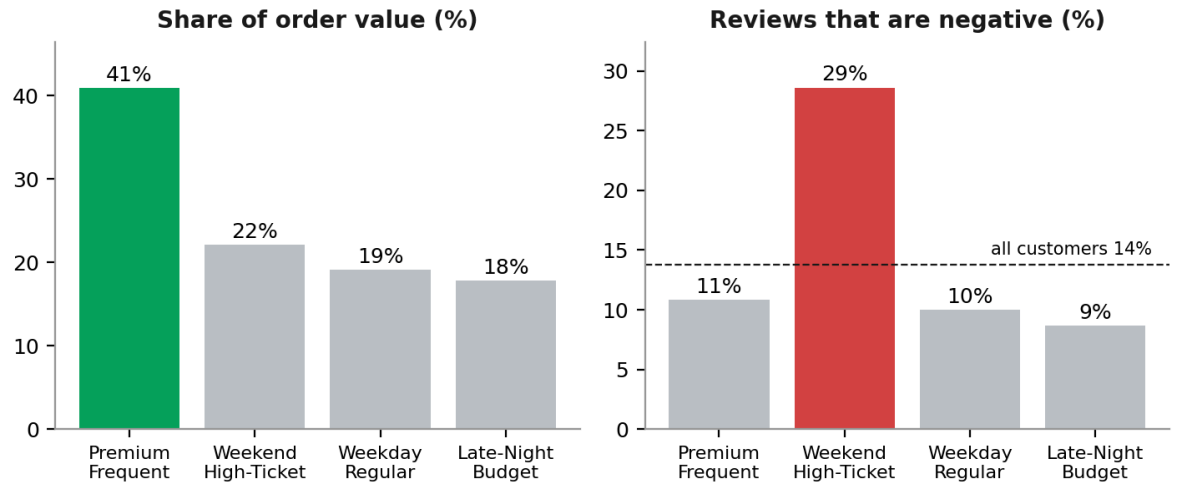


Figure 1. Share of delivered order value, and share of reviews that are negative, by customer segment.

Segment	Customers	Order value	Avg order (₹)	Orders each	What stands out
Premium Frequent Diners	156	₹14.6 lakh (41%)	780	12.0	Most orders and highest lifetime spend (₹9,385 each)

Segment	Customers	Order value	Avg order (₹)	Orders each	What stands out
Weekend High-Ticket	119	₹7.9 lakh (22%)	887	7.6	Biggest baskets; lowest satisfaction (avg rating 3.3)
Weekday Regulars	113	₹6.8 lakh (19%)	853	7.0	Large baskets, satisfied, rarely order at weekends
Late-Night Budget	112	₹6.4 lakh (18%)	755	7.5	Smallest baskets; 48% of orders after 10 PM

Weekend High-Ticket customers write 41% of all negative reviews while making up 24% of customers. If their dissatisfaction matched the other segments, there would be roughly 70 fewer negative reviews.

Which restaurant segments require intervention?

Answer: High-Demand Slow-Preparation restaurants first, then Fast-Prep Lower-Rated restaurants.

Segment	Count	Share of orders	Avg rating	Prep time (min)	Avg delivery (min)	Over 60 min
High-Demand Slow-Preparation	**20**	**42%**	**4.2**	**33.7**	**68.4**	**68% of orders**
Fast-Prep Lower-Rated	19	39%	3.6	15.6	51.0	28% of orders
Highly Rated Niche	11	19%	4.2	24.3	58.7	47% of orders

- **Slow-preparation group:** the biggest group by volume and revenue (₹15.3 lakh), with the longest deliveries. Since each minute in the kitchen adds about a minute to delivery, bringing their prep time down to the niche group's level (about 24 minutes) would save roughly 9 minutes per order.
- **Lower-rated group:** fast kitchens and quick deliveries, but the lowest star rating (3.6) and the highest cancellation rate of the three groups (9.4%). The issue here is food quality, not speed.
- **Watchlist:** five individual restaurants have negative-review rates of 24% to 29% against 14% overall (R041, R002, R006, R045, R018, each with 31 to 46 reviews). Worth a closer look, though the sample sizes are modest.

What are the biggest causes of negative customer experiences?

Answer: Slow or late delivery (35%) and food quality problems (28%), with a large share of unspecific dissatisfaction (37%).

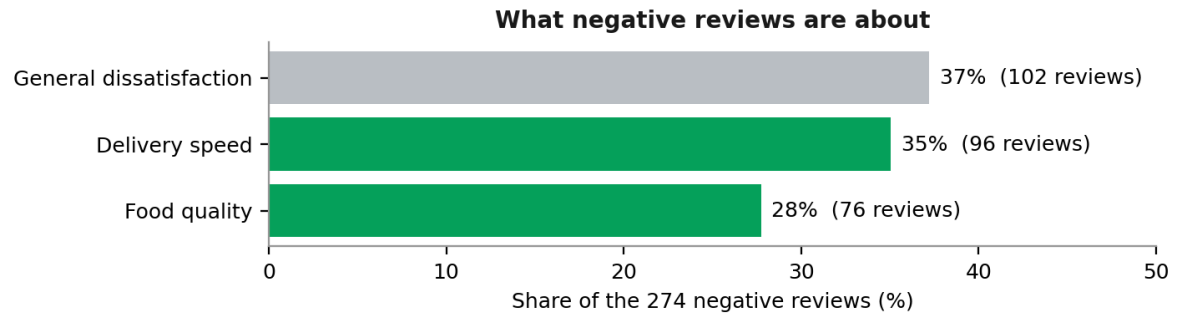


Figure 2. Themes in the 274 negative (1 to 2 star) reviews.

- Delivery speed: "Delivery was slow" (69) and "Very late delivery and cold food" (27).

- Food quality: "Food quality was disappointing" (49) and "Food arrived damaged" (27).
- General: "Could be much better" (66) and "Terrible experience" (36).

The strongest predictor of a negative review is who the customer is, not what happened on the order. The Weekend High-Ticket segment is negative 29% of the time versus 9% to 11% elsewhere. Traffic, weather, distance and measured delivery time show little link to negative reviews: orders that received negative reviews were delivered in 59 minutes on average, the same as positive ones (60). This suggests expectations differ between segments, or that review wording is not tied closely to the actual delivery.

Which factors drive delivery delays?

Answer: **Distance and kitchen preparation time, followed by heavy traffic and rain.**

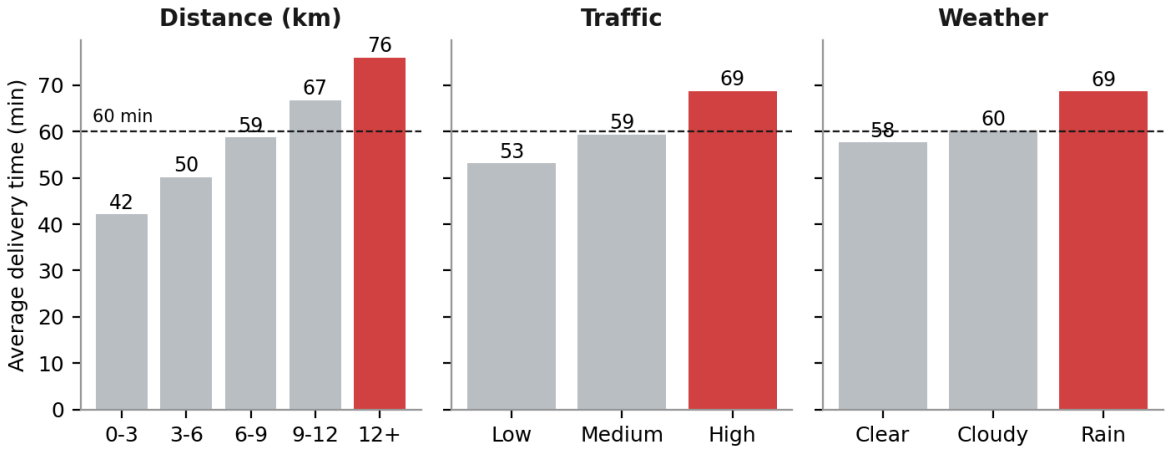


Figure 3. Average delivery time by distance, traffic and weather. The dashed line marks 60 minutes.

Factor	Effect on delivery time	Evidence
Distance	About +3 min per km	Orders over 12 km average 76 min; 88% exceed an hour. Under 3 km: 42 min.
Kitchen preparation	About +1 min per prep minute	Orders from kitchens above 35 min prep average 73 min.
Heavy traffic	About +16 min vs light traffic	69% of heavy-traffic orders exceed an hour (34% in light traffic).
Rain	About +12 min vs clear weather	70% of rainy-day orders exceed an hour (44% in clear weather).
Rain and heavy traffic together	Worst case	Average 78 min, against 51 min in clear weather with light traffic.
Order size, driver experience, vehicle type	No meaningful effect	Bike, motorcycle and scooter deliveries all average about 60 min.

Pune has the slowest deliveries (63 min average; 57% over an hour) and Bengaluru the fastest (56 min). Weekends are no slower than weekdays.

What factors influence tip amounts?

Answer: **Mainly order value. Nothing operational (speed, weather, traffic, payment method) shows an effect.**

- The average tip is ₹34.60. A quarter of delivered orders get no tip.

- Tips fall into four equal groups: 0%, 2%, 5% and 10% of the order total, each about 25% of orders. Which group an order lands in is unrelated to delivery time, weather, traffic, distance, payment method, city, customer segment or star rating.
- Tip size in rupees rises with basket size (average ₹12 on the smallest quarter of orders versus ₹63 on the largest), simply because the percentage applies to a bigger total.

Implication: faster or better service is unlikely to lift tips in this data. Driver incentives should not be built on the assumption that tips reward good service. A dedicated tip prediction model was not built, since delivery time was chosen as the primary prediction target.

When should additional delivery capacity be deployed?

Answer: When conditions are bad (rain, heavy traffic) and for long-distance orders, rather than at fixed hours of the day.

Demand does not follow a dependable daily or weekly rhythm in this data (see next question), so clock-based rostering would add cost without a measurable benefit. The data does show where delivery time breaks down: rain and heavy traffic add 12 to 16 minutes, and orders beyond 9 km (43% of all orders) have a 68% to 88% chance of passing the one-hour mark. Extra riders, tighter dispatch radius or a stretched delivery promise are most useful in those situations.

Which locations and time periods are likely to see demand spikes?

Answer: No reliable time-based spikes. Location matters more: Delhi and Mumbai carry most of the volume.

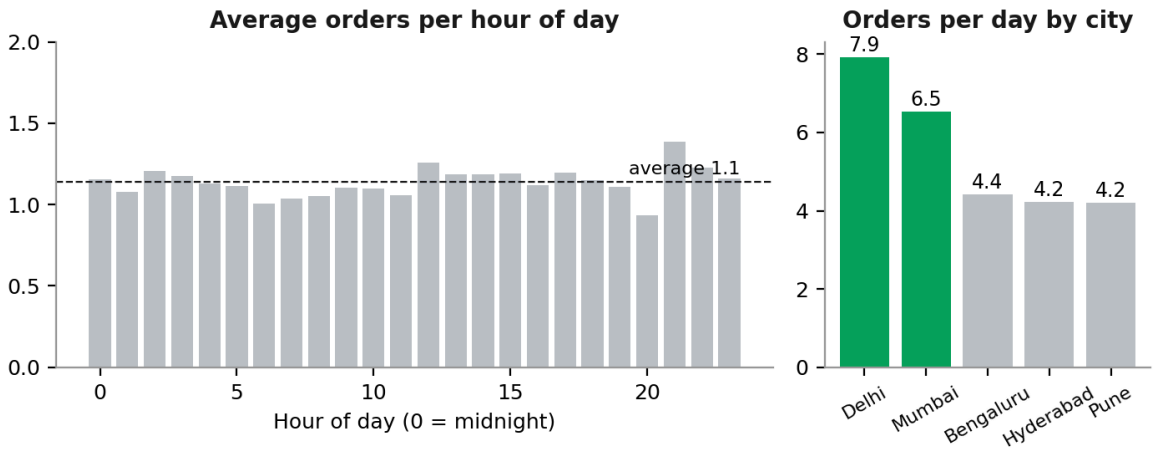


Figure 4. Average orders per hour of day, and average orders per day by city of the restaurant.

- Demand averages about 27 orders a day (1.1 an hour). The busiest hour of the day (9 PM, 1.4 orders) is only about 20% above average, and the hourly differences are within normal random variation.
- Saturday is the busiest day (29 orders a day against 26 on Mondays), but the gap is not statistically meaningful. The most active individual slots are Saturday 10 PM and Thursday 9 PM, at under 2 orders per hour, which is too few to plan around.
- Delhi (7.9 orders a day) and Mumbai (6.5) account for 53% of orders. Bengaluru, Hyderabad and Pune each run at about 4 a day.
- Underlying growth is steady: monthly orders rose from 784 in April to 885 in September (+13%).

The example spike most teams expect, a weekend dinner rush, is not present in this data: weekend dinner-hour orders (4.7 per evening) are almost the same as weekday ones (4.5).

4. Recommendations

Ordered by expected impact. Each is tied directly to evidence from the analysis.

#	Recommendation	Evidence	Expected result
1	Launch a retention programme for Weekend High-Ticket customers: priority support, proactive follow-up after weekend orders, and a root-cause review of their complaints.	29% of their reviews are negative versus 10% elsewhere. They are 22% of order value and have the largest baskets (₹887).	Closing half the gap would remove about 35 negative reviews per period and protect ₹7.9 lakh of order value.
2	Set a preparation-time target for the 20 slow-preparation restaurants (aim for under 25 minutes) and track it weekly.	They handle 42% of orders and deliver in 68 min on average. Each prep minute adds about one delivery minute.	Up to 9 minutes faster per order for these restaurants, cutting the share of orders over an hour well below 68%.
3	Trigger extra rider capacity and show a longer delivery promise during rain and heavy traffic, rather than at fixed hours.	Rain adds about 12 minutes and heavy traffic about 16. Both together average 78 minutes versus 51.	Fewer late orders on bad-condition days and more honest customer expectations.
4	Tighten dispatch for long-distance orders: assign the nearest rider, or apply a distance-based surcharge or delivery promise above 9 km.	43% of orders travel over 9 km; 68% to 88% of these take over an hour. Each km adds about 3 minutes.	Shorter average delivery times and fewer late-delivery complaints.
5	Run a food-quality improvement plan for the 19 Fast-Prep Lower-Rated restaurants, starting with the five watchlist restaurants.	Lowest rating (3.6), highest cancellation rate (9.4%). Quality complaints are 28% of negative reviews.	Higher ratings and fewer failed orders without slowing delivery.
6	Use the delivery time model to show customers a predicted arrival time at checkout.	Typical prediction error is 3.2 minutes, against 14 minutes without the model.	More accurate promises and fewer "slow delivery" complaints (35% of negative reviews).
7	Prioritise operations in Delhi and Mumbai, and investigate Pune delivery times.	Delhi and Mumbai are 53% of orders. Pune is the slowest city (63 min; 57% over an hour).	Effort focused where volume and delay are highest.
8	Investigate cancellations and failed orders (about 1 in 9 orders).	445 cancelled and 145 failed orders worth ₹4.6 lakh; rate is highest in Pune (13.5%).	Recovered order value and fewer disappointed customers.

5. Model limitations

- **Synthetic data.** The dataset was generated for this project. Several relationships (for example, tips independent of service, or negative reviews unrelated to delivery time) are unusual compared with real marketplaces and should not be assumed to hold in live operations.
- **Small volumes.** About 27 orders a day and about 1 order per hour make hourly forecasting inherently noisy. A forecast can describe the average level but cannot pinpoint individual busy hours.
- **Short history.** Six months of data cannot show yearly seasonality, holidays or festivals.
- **Weak customer segmentation.** The customer groups overlap (separation score 0.16), so individual customers can sit between segments. Restaurant segments are based on only 50 restaurants.
- **Sentiment model scores are inflated.** Reviews use a small set of repeated phrases and the sentiment label comes from the star rating itself. Real-world reviews would be harder, and accuracy would be lower. Only 45% of delivered orders have a review.
- **Delivery model uses only a handful of inputs.** It does not include rider availability, real-time road conditions or restaurant queue length, and it describes patterns, not proven cause and effect.

- **No location detail below city level.** Demand cannot be broken down by neighbourhood or delivery zone.

6. Future improvement opportunities

- **Connect to live data** and re-validate every finding in this report before acting on it.
- **Add richer inputs:** rider supply by hour, restaurant order queues, live traffic and weather forecasts, neighbourhood-level order locations.
- **Build a tip prediction model** once real tipping data is available, to test whether service quality drives tips.
- **Move to richer sentiment analysis:** use longer, free-text reviews and a language model that can pick up topics such as cold food or rider behaviour.
- **Predict churn and lifetime value** by segment, to quantify the cost of the Weekend High-Ticket dissatisfaction.
- **Forecast by city and by delivery zone** with more history, and add holidays and promotions as inputs.
- **Automate and monitor:** schedule the pipeline to retrain regularly, track model accuracy over time and alert the team when it drifts.
- **Run controlled pilots** for the recommendations above (for example, a retention offer for one group of customers) to measure real impact.